Name: _____

TOPIC TEST 1

Chapter 1: Three-dimensional vectors

- Time allowed: 45 minutes
- CAS calculators may be used.
- Part A: 5 multiple-choice questions (5 marks).
- Part B: 6 short-answer questions (44 marks).
- Part C: 2 analysis questions (11 marks).

Part A

Multiple-choice questions

5 multiple-choice questions
1 mark each: 5 marks
Circle the correct answer.

1 $Q(2, -3, -5)$ and which of the following points are on a line parallel with the z-axis?

A $(-2, 3, -5)$

B $(2, 3, -5)$

C $(2, -3, 6)$

D $(4, -2, -5)$

E $(5, -2, -3)$

[1 mark]

2 The distance between $P(a, b, c)$ and $Q(d, b, c)$ is:

A $|a - b|$

B $|c - a|$

C $|b - c|$

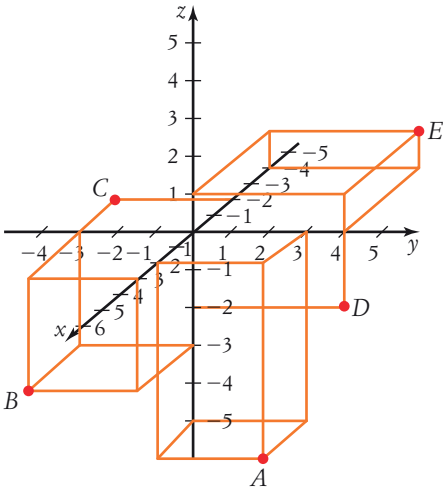
D $|a - d|$

E $|a - c|$

[1 mark]

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Use the diagram below to answer Questions 3–5.



3 The point lying on the *yz*-plane is:

A *A*

B *B*

C *C*

D *D*

E *E*

[1 mark]

4 The point *C* is best described as lying on the:

A *yz*-plane

B *y*-plane

C *xz*-plane

D *x*-plane

E *xy*-plane

[1 mark]

5 The points lying on a plane parallel to the *xz*-plane are:

A *E* and *C*

B *A* and *D*

C *B* and *D*

D *B* and *C*

E *A* and *E*

[1 mark]

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Part B

Short-answer questions

6 short-answer questions
44 marks
Show your working where appropriate.

1 Find the radius and centre of the sphere
$x^2 + y^2 + z^2 - 4x + 8y - 6z - 20 = 0$

[6 marks]

2 Given $\mathbf{m} = (-1, 3, 4)$, $\mathbf{n} = (5, -2, 6)$, and $A = (4, 3, -7)$ and $B = (-3, -1, 4)$:

a find $-3\mathbf{m} + 4\mathbf{n}$

b show that $(\mathbf{m} - \mathbf{n})$ is parallel to $(\mathbf{n} - \mathbf{m})$

c find $\mathbf{AB}$ and $|\mathbf{AB}|$

d find $\hat{\mathbf{m}}$

[10 marks]

3 **a** Show that the vectors $(-1, -3, 4)$ and $(6, 2, 3)$ are perpendicular.

b Find the projection of $\mathbf{m} = (-2, 1, -5)$ on $\mathbf{n} = (1, -3, 4)$.

c Find the angle between $\mathbf{p} = 2\mathbf{i} + 5\mathbf{j} - 3\mathbf{k}$ and $\mathbf{q} = \mathbf{i} - 4\mathbf{j} - 3\mathbf{k}$.

[10 marks]

4 **a** Find the vector product of vectors of magnitudes 15 and 9 if the angle between them is 75°.

b Find the vector product of $\mathbf{u} = (3, -1, 5)$ and $\mathbf{v} = (-4, 2, -3)$.

[7 marks]

5 **a** Find the Cartesian form of the equation of the straight line which passes through the points $(2, 5, -3)$ and $(1, 3, -6)$.

b Calculate the equivalent vector equation for the line.

[6 marks]

6 Find the equation of the plane that contains the points $P(3, -1, 4)$, $Q(6, 2, 5)$ and $R(-2, 1, 3)$.

[5 marks]

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Part C

Analysis questions

2 analysis questions
11 marks
Show your working where appropriate.

1 Prove the parallelogram law which states:

The sum of the squares of lengths of both diagonals of a parallelogram is equal to the sum of the squares of the lengths of all four sides.

[7 marks]

2 Find the equation of the plane that is equidistant from the points $A(3, 2, 1)$ and $B(-3, -2, -1)$ (that is, every point on the plane has the same distance from the two given points).

[4 marks]

[Total marks: 60]

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Answers

Part A – Multiple-choice questions [1 mark each]

1 Points on a line parallel to the z-axis will have x- and y-coordinates the same but different z-coordinates.

$\therefore$ **C** **[1 mark]**

2 P and Q have different x-coordinates but the y- and z-coordinates are the same.

The points are on a line parallel to the x-axis.

The distance between P and Q will be the absolute value of the difference between the x-coordinates i.e. $|a - d|$

$\therefore$ **D** **[1 mark]**

3 Any point on the yz-plane will have an x-coordinate of 0.

$D = (0, 4, 0)$.

$\therefore$ **D** **[1 mark]**

4 $C = (-2, -3, 0)$.

The z-coordinate is 0, C lies on the xy-plane.

$\therefore$ **E** **[1 mark]**

5 Any points lying on a plane parallel to the xz-plane will have the same y-coordinate.

$B = (4, -3, -3)$ and $C = (-2, -3, 0)$

$\therefore$ **D** **[1 mark]**

Part B – Short-answer questions

1 $x^2 + y^2 + z^2 - 4x + 8y - 6z - 20 = 0$

$x^2 + y^2 + z^2 - 4x + 8y - 6z = 20$

$(x^2 - 4x) + (y^2 + 8y) + (z^2 - 6z) = 20$ **[1 mark]**

$(x^2 - 4x + 4) + (y^2 + 8y + 16) + (z^2 - 6z + 9) = 4 + 16 + 9 + 20$ **[1 mark]**

Factorise the squares. $(x - 2)^2 + (y + 4)^2 + (z - 3)^2 = 49$ **[1 mark]**

$(x - 2)^2 + (y + 4)^2 + (z - 3)^2 = 7^2$ **[1 mark]**

Compare with $(x - h)^2 + (y - k)^2 + (z - l)^2 = r^2$

Centre $= (2, -4, 3)$ **[1 mark]**

Radius $= 7$ **[1 mark]**

[6 marks altogether]

2 a $-3\mathbf{m} + 4\mathbf{n} = -3(-1, 3, 4) + 4(5, -2, 6)$ **[1 mark]**

$= (3, -9, -12) + (20, -8, 24)$

$= (23, -17, 12)$ **[1 mark]**

b $\mathbf{m} - \mathbf{n} = (-1, 3, 4) - (5, -2, 6)$

$= (-6, 5, -2)$ **[1 mark]**

$\mathbf{n} - \mathbf{m} = (5, -2, 6) - (-1, 3, 4)$

$= (6, -5, 2)$

$= -1(-6, 5, -2)$ **[1 mark]**

$= -1(\mathbf{m} - \mathbf{n})$

Thus $(\mathbf{m} - \mathbf{n})$ is parallel to $(\mathbf{n} - \mathbf{m})$ **[1 mark]**

c $\mathbf{AB} = (x_2 - x_1, y_2 - y_1, z_2 - z_1)$

$= (-3 - 4, -1 - 3, 4 + 7)$

$= (-7, -4, 11)$ **[1 mark]**

$|\mathbf{AB}| = \sqrt{a_1^2 + a_2^2 + a_3^2}$

$= \sqrt{(-7)^2 + (-4)^2 + 11^2}$

$= \sqrt{186}$ **[1 mark]**

d $\hat{\mathbf{m}} = \dfrac{1}{|\mathbf{m}|}\mathbf{m}$ **[1 mark]**

$= \dfrac{1}{\sqrt{(-1)^2 + 3^2 + 4^2}}(-1,3,4)$ **[1 mark]**

$= \dfrac{1}{\sqrt{26}}(-1,3,4)$

$= \dfrac{\sqrt{26}}{26}(-1,3,4)$ **[1 mark]**

[10 marks altogether]

3 a $(-1, -3, 4) \cdot (6, 2, 3) = -6 - 6 + 12$

$= 0$ **[1 mark]**

Since the scalar product is zero and the vectors are not zero vectors, the vectors are perpendicular. **[1 mark]**

b $p_n\,\mathbf{m} = \mathbf{m} \cdot \hat{\mathbf{n}}$ **[1 mark]**

$\hat{\mathbf{n}} = \dfrac{1}{|\mathbf{n}|}\mathbf{n}$

$= \dfrac{1}{\sqrt{1^2 + (-3)^2 + 4^2}}(1, -3, 4)$

$= \dfrac{1}{\sqrt{26}}(1, -3, 4)$ **[1 mark]**

$p_n\,\mathbf{m} = (-2, 1, -5) \cdot \dfrac{1}{\sqrt{26}}(1, -3, 4)$

$= \dfrac{1}{\sqrt{26}}(-2 - 3 - 20)$ **[1 mark]**

$= -\dfrac{25}{\sqrt{26}} \approx -4.9$ **[1 mark]**

c $\cos(\theta) = \dfrac{\mathbf{p} \cdot \mathbf{q}}{|\mathbf{p}||\mathbf{q}|}$ **[1 mark]**

$= \dfrac{2 - 20 + 9}{\sqrt{2^2 + 5^2 + (-3)^2} \times \sqrt{1^2 + (-4)^2 + (-3)^2}}$ **[1 mark]**

$= \dfrac{-9}{\sqrt{38} \times \sqrt{26}}$

$= \dfrac{-9}{2\sqrt{247}}$ **[1 mark]**

$\theta \approx 106.6°$ **[1 mark]**

[10 marks altogether]

4 a $|\mathbf{v}| = |\mathbf{v}_1|\,|\mathbf{v}_2| \sin(\theta)$ **[1 mark]**

$= 15 \times 9 \times \sin(75°)$ **[1 mark]**

≈ 130.4 **[1 mark]**

The vector product is about 130 in a direction perpendicular to the plane of the original vectors. **[1 mark]**

b $\mathbf{u} = 3\mathbf{i} - \mathbf{j} + 5\mathbf{k}, \mathbf{v} = -4\mathbf{i} + 2\mathbf{j} - 3\mathbf{k}$ **[1 mark]**

$\mathbf{u} \times \mathbf{v} = (-1 \times -3 - 5 \times 2)\mathbf{i} - (3 \times -3 - 5 \times -4)\mathbf{j} + (3 \times 2 - -1 \times -4)\mathbf{k}$ **[1 mark]**

$= -7\mathbf{i} - 11\mathbf{j} + 2\mathbf{k}$

$= (-7, -11, 2)$ **[1 mark]**

[7 marks altogether]

5 a Let $A(a_1, a_2, a_3) = (2, 5, -3)$

and $B(b_1, b_2, b_3) = (1, 3, -6)$ **[1 mark]**

$\dfrac{x - a_1}{b_1 - a_1} = \dfrac{y - a_2}{b_2 - a_2} = \dfrac{z - a_3}{b_3 - a_3}$

$\dfrac{x - 2}{1 - 2} = \dfrac{y - 5}{3 - 5} = \dfrac{z - (-3)}{-6 - (-3)}$ **[1 mark]**

$\dfrac{x - 2}{-1} = \dfrac{y - 5}{-2} = \dfrac{z + 3}{-3}$ **[1 mark]**

b Compare $\dfrac{x - 2}{-1} = \dfrac{y - 5}{-2} = \dfrac{z + 3}{-3}$ with $\dfrac{x - a_1}{u_1} = \dfrac{y - a_2}{u_2} = \dfrac{z - a_3}{u_3}$

$a_1 = 2, a_2 = 5, a_3 = -3$

and $u_1 = -1, u_2 = -2, u_3 = -3$ **[1 mark]**

$\mathbf{p} = \mathbf{a} + t\mathbf{u}$

$= a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k} + t(u_1\mathbf{i} + u_2\mathbf{j} + u_3\mathbf{k})$ **[1 mark]**

$\mathbf{p} = 2\mathbf{i} + 5\mathbf{j} - 3\mathbf{k} + t(-\mathbf{i} - 2\mathbf{j} - 3\mathbf{k})$ **[1 mark]**

[6 marks altogether]

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6 $\mathbf{PR} = (2 - 3, 1 - (-1), 3 - 4) = (-5, 2, -1)$

$\mathbf{PQ} = (6 - 3, 2 - (-1), 5 - 4) = (3, 3, 1)$ **[1 mark]**

$\mathbf{PR} \times \mathbf{PQ}$ is normal to the plane containing $\mathbf{PR}$ and $\mathbf{PQ}$ **[1 mark]**

$\mathbf{PR} \times \mathbf{PQ} = (2 \times 1 - (-1) \times 3)\mathbf{i} - (-5 \times 1 - (-1) \times 3)\mathbf{j} + (-5 \times 3 - 2 \times 3)\mathbf{k}$

$= 5\mathbf{i} + 2\mathbf{j} - 21\mathbf{k} = \mathbf{n}$ **[1 mark]**

Cartesian equation of the plane is: $a_1(x - x_1) + a_2(y - y_1) + a_3(z - z_1) = 0$

Let $P(3, -1, 4)$, and $\mathbf{n} = (5, 2, -21)$ **[1 mark]**

$5(x - 3) + 2(y + 1) - 21(z - 4) = 0$

$5x - 15 + 2y + 2 - 21z + 84 = 0$

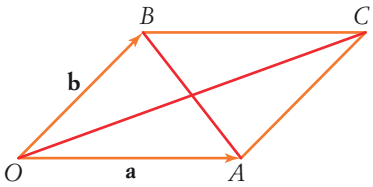
$5x + 2y - 21z + 71 = 0$ is the equation of the plane **[1 mark]**

[5 marks altogether]

Part C – Analysis questions

1 Draw a sketch.

Label the vertices O, A, B and C.



[1 mark]

$\mathbf{OA} = \mathbf{a}$

$\mathbf{OB} = \mathbf{b}$

$\mathbf{OC} = \mathbf{a} + \mathbf{b}$

$\mathbf{AB} = \mathbf{a} - \mathbf{b}$ **[1 mark]**

$|\mathbf{a}|^2 = \mathbf{a} \cdot \mathbf{a}$

$|\mathbf{b}|^2 = \mathbf{b} \cdot \mathbf{b}$ **[1 mark]**

$|\mathbf{a} + \mathbf{b}|^2 = (\mathbf{a} + \mathbf{b}) \cdot (\mathbf{a} + \mathbf{b})$

$|\mathbf{a} - \mathbf{b}|^2 = (\mathbf{a} - \mathbf{b}) \cdot (\mathbf{a} - \mathbf{b})$ **[1 mark]**

The squares of the lengths of the diagonals

$= |\mathbf{a} + \mathbf{b}|^2 + |\mathbf{a} - \mathbf{b}|^2$

$= (\mathbf{a} + \mathbf{b}) \cdot (\mathbf{a} + \mathbf{b}) + (\mathbf{a} - \mathbf{b}) \cdot (\mathbf{a} - \mathbf{b})$

$= [\mathbf{a} \cdot \mathbf{a} + 2(\mathbf{a} \cdot \mathbf{b}) + \mathbf{b} \cdot \mathbf{b}] + [\mathbf{a} \cdot \mathbf{a} - 2(\mathbf{a} \cdot \mathbf{b}) + \mathbf{b} \cdot \mathbf{b}]$

$= 2(\mathbf{a} \cdot \mathbf{a}) + 2(\mathbf{b} \cdot \mathbf{b})$

$= 2|\mathbf{a}|^2 + 2|\mathbf{b}|^2$ **[1 mark]**

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The squares of the lengths of the four sides

$= |\mathbf{a}|^2 + |\mathbf{a}|^2 + |\mathbf{b}|^2 + |\mathbf{b}|^2$

$= 2|\mathbf{a}|^2 + 2|\mathbf{b}|^2$ **[1 mark]**

= The squares of the lengths of the diagonals **[1 mark]**

QED

[7 marks altogether]

2 The midpoint of the points A and B is the point

$C = 0.5[(3, 2, 1) + (-3, -2, -1)] = (0, 0, 0)$ **[1 mark]**

A normal vector to the plane is given by

$\mathbf{AB} = (3, 2, 1) - (-3, -2, -1) = (6, 4, 2)$ **[1 mark]**

So, the equation of the plane is $6(x - 0) + 4(y - 0) + 2(z - 0) = 0$ **[1 mark]**

$3x + 2y + z = 0$ **[1 mark]**

[4 marks altogether]

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